

Experiment 9: Rotation of an Image (Clockwise and Counter Clockwise)

Aim

To read an image using Python and rotate it in clockwise and counter clockwise directions using OpenCV.

Algorithm

1. Import the OpenCV library.
2. Read the input image.
3. Rotate the image clockwise.
4. Rotate the image counter clockwise.
5. Display the original and rotated images.
6. Close all display windows.

Python Code

```
import cv2

# Read the image
img = cv2.imread("tree.jpg")

# Check whether image is loaded
if img is None:
    print("Error: Image not found!")
    exit()

# Rotate the image 90 degrees clockwise
clockwise = cv2.rotate(img, cv2.ROTATE_90_CLOCKWISE)

# Rotate the image 90 degrees counter clockwise
counter_clockwise = cv2.rotate(img, cv2.ROTATE_90_COUNTERCLOCKWISE)

# Display all images
cv2.imshow("Original Image", img)
cv2.imshow("Clockwise Rotation", clockwise)
cv2.imshow("Counter Clockwise Rotation", counter_clockwise)

# Wait until any key is pressed
cv2.waitKey(0)
```

```
# Close all windows  
cv2.destroyAllWindows()
```

Input



Output

Clockwise Rotation



Counter Clockwise Rotation



Result

The image was successfully rotated in clockwise and counter clockwise directions using Python and OpenCV.